Case 35: Midflame wind-speed Rothermel sweep

1 Verification purpose

This case verifies ELMFIRE's Rothermel wind response for FBFM2, FBFM9, GR2, TU5, and TL8 over 0, 1, 2, 4, 6, 8, and 12 mph midflame. It isolates the midflame wind factor while separately accounting for ELMFIRE's public 20-ft wind input and unsheltered wind-adjustment factor (WAF). It can expose errors in wind units, WAF inversion, exponentiation, fuel-dependent coefficients, or wind limiting. The question is whether every simulation condition agrees with the independent implementation-limited rate of spread (ROS) within 2%, reaction intensity within 1%, and time of arrival (TOA)-derived ROS within 5%.

2 Mathematical formulation and numerical implementation

Rothermel wind enhancement is

$$R = R_0(1 + \phi_w), \qquad \phi_w = CU^B(\beta/\beta_{op})^{-E}, \tag{1}$$

with U in ft min^{-1} at midflame. Coefficients are $C = 7.47e^{-0.133\sigma^{0.55}}$, $B = 0.02526\sigma^{0.54}$, and $E = 0.715e^{-0.000359\sigma}$. The reference is Andrews (2018), RMRS-GTR-371. The inspected ELMFIRE implementation limits U to $0.9I_R$ in native units. ELMFIRE inputs 20-ft wind in mph, so preprocessing divides each target midflame speed by the independently computed unsheltered WAF.

3 Derivation of expected behavior

Below the limit, $(R/R_0 - 1)$ follows the fuel-specific power law in U; once the limit is reached, the wind contribution is constant. The case-specific reference calculation computes R_0, I_R, WAF, ϕ_w, limit location, and final ROS from the paper equations and the documented ELMFIRE limiting policy. This separates paper-equation behavior from the implementation's explicit cap.

The wind series uses the Rothermel wind factor

$$\phi_w = C\,U_{mf}^B, \qquad R = 0.3048R_0^{(\text{ft/min})}(1 + \phi_w),$$

with U_{mf} in ft min^{-1} and fuel-specific B, C. For a concrete generated substitution, Fuel Model 2 at $U_{mf} = 4$ mph $= 352$ ft min^{-1} has $C = 0.0002125869$, $B = 1.8304427$, and

$$\phi_w = 0.0002125869(352)^{1.8304427} = 9.746235,$$

$$R_{\text{ref}} = 0.3048(2.702741)(1 + 9.746235) = 8.852700 \text{ m min}^{-1}, \qquad I_{R,\text{ref}} = 675.184 \text{ kW m}^{-2}.$$

The same substitution is applied to every generated fuel/wind pair ($0, 1, 2, 4, 6, 8, 12$ mph); the 20-ft raster value is $U_{20} = U_{mf}/WAF$, e.g. $4/0.362674 = 11.02919$ mph. For each simulation condition, $e_R = |R_{\text{max}} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and e_I is analogous. A TOA regression $t = a + bx$ gives $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregated tests substitute max $e_R \le 0.02$, max $e_I \le 0.01$, max $e_{\text{TOA}} \le 0.05$, and min $R^2 \ge 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for the final direct ROS, 0.01 for reaction intensity, and 0.05 for TOA-derived ROS. The minimum TOA regression R^2 is 0.995. Direct pointwise comparison is more diagnostic than monotonicity alone because it detects an incorrect exponent or premature plateau even when the curve still increases. All 35 fuel–wind combinations are required.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE's stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. Each homogeneous 1400 m square simulation condition uses 10 m cells and a finite ignition strip whose east face is planar at $x = -400$ m within the interior analysis corridor. The strip ends inside the lateral boundaries so it remains an active fire rather than an edge-to-edge contained front. Slope is zero, dead moisture is 5/7/9%, and live moisture is 90%. The wind is from the west. Crown fire, spotting, buildings, fluctuations, acceleration, and Monte Carlo variation are disabled. Curves should increase as U^B and may flatten at fuel-dependent limits.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel sweep	FBFM2, FBFM9, GR2, TU5, and TL8	code/name	Fuel-dependent response
Wind sweep	0, 1, 2, 4, 6, 8, and 12 mph midflame	mph	Independent midflame target
Moisture/slope	dead 5/7/9, live 90/90; 0	percent / degrees	Fixed damping and terrain
Timing	generated whole-second step and exact-multiple stop	s	CFL-safe terminal evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

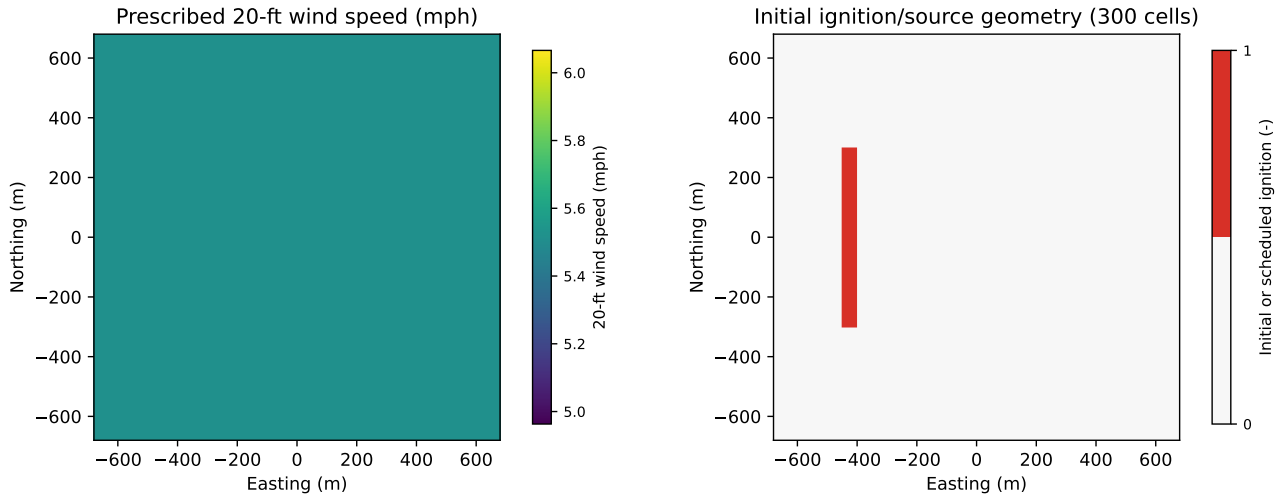


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 35 of 35 required simulation conditions and reports **PASS**.

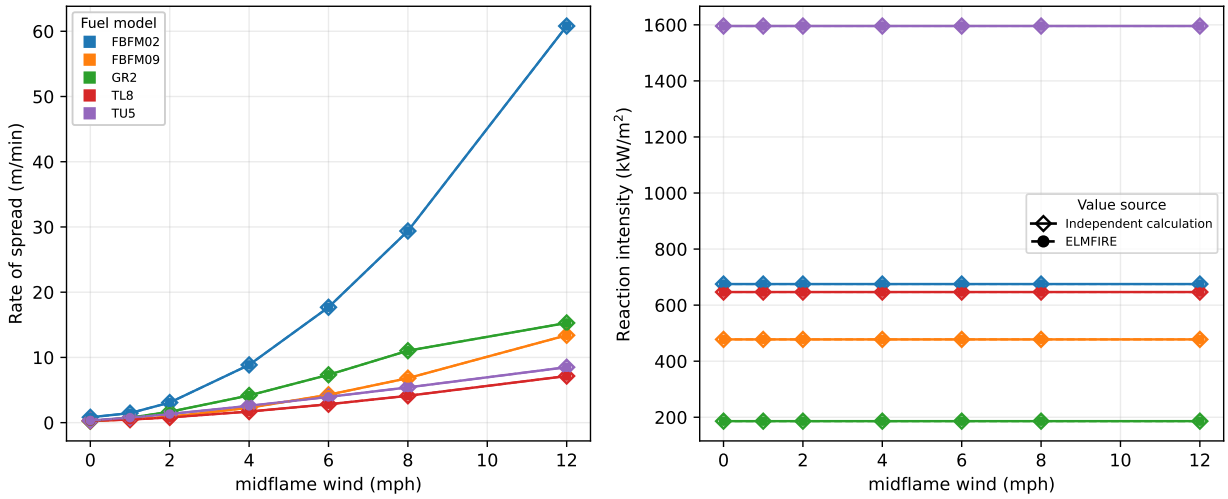


Figure 2: Independent calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines). Each fuel-model pair uses the same color. Target values are midflame wind; namelists contain the corresponding 20-ft wind.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	1.17774e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.93279e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.028404 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999958 dimensionless	PASS

The authoritative overall decision is **PASS**.

8 Technical interpretation

A common scale error across fuels suggests mph, ft/min, or WAF conversion. Fuel-dependent curvature errors suggest B, C, E, or packing-ratio handling. A wrong plateau indicates a changed or incorrect wind-limit policy. Correct direct ROS but incorrect TOA points to transport rather than the Rothermel calculation.

9 Scope and limitations

The case verifies steady, aligned wind and the currently selected original wind limit. If a future ELM-FIRE release intentionally adopts a corrected or optional limit, the scientific reference calculation must be reviewed rather than changing only the namelist. Canopy-dependent WAF is tested in CASE42.